

./02-architecture/DOC-04-System-Architecture.md

DOC-04: System Architecture Document (SAD)

Version: 1.0 | **Status:** APPROVED | **Date:** 2026-08-27 **Purpose:** Define the high-level system architecture and constraints. **Owner:** Senior Software Architect

Architecture Overview

- **Pattern:** Modular Monolith + API-first.
- **Frontend:** Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, TanStack Query.
- **Backend:** NestJS, Node.js, TypeScript.
- **Database:** PostgreSQL with Prisma ORM.

Logical Architecture

- **Clients:** Next.js Web (Future: Flutter).
- **API Gateway/Layer:** REST API via NestJS.
- **Infrastructure Services:** Redis (for rate limiting, queues, session data), S3-compatible Object Storage.

Architectural Principles & Rules

1. TypeScript everywhere.
2. Modular Monolith. No premature microservices.
3. Multi-tenant by design. Application-level tenant isolation.
4. Backend is the source of truth for authorization.
5. Controllers remain thin; Services contain business logic.
6. DTO validation is mandatory.

```
graph TD
  Client[Next.js Web Client] --> REST_API[NestJS REST API]
  REST_API --> Modules[Business Modules]
  Modules --> Prisma[Prisma ORM]
  Prisma --> DB[(PostgreSQL)]
  REST_API --> Redis[(Redis Cache/Queue)]
  REST_API --> S3[(S3 Object Storage)]
```